Mathematics for Biomedical Engineering

L10:

Function of Multiple Variables

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Key Learning Objectives

- understand how function can depend on more than one variable
- understand how function of more than one variable can be represented in space and in different coordinate systems
- be able to differentiate a function of more than one variable
- be able to integrate functions over more than one variable
- be able to perform differentiation and integration of function of more than one variable in different coordinate systems

Section 1 of this lecture is based-on Chapter 22 from Croft & Davidson, Mathematics for Engineers.
Section 2 of this lecture is based-on Chapters 22 & 23 from Stroud, Engineering Mathematics.

Introduction

We are used to thinking about a function like a mathematical rule that operates upon an input to produce a single output.

$$y = f(x), \tag{1}$$

where x is the *independent variable*, y is the *dependent variable*.

Now we consider a function having *two* independent variables, but still producing a single output (the dependent variable)

$$z = f(x, y), \tag{2}$$

where x and y are the independent variables and z is the dependent variable.

Function of two variables

Consider the function

$$f(x, y) = 3x + 4y \tag{3}$$

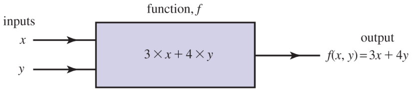

Figure: A block diagram of the function $f(x, y) = 3x + 4y$.

The inputs to the function are x and y, which are also known as **arguments**.

Function of two variables

A function of two variables is a rule that produces a single output when values of two independent variables are chosen.

Example: The volume of a cylinder is given by $V(r, h) = \pi r^2 h$. Determine V if $r = 2$, $h = 3$. Are r and h independently related?

Contour plots of function of two variables

A **contour plot** is a useful way of representing a function of two independent variables $z = f(x, y)$.

Contour maps draw lines or contours which have the same value z.

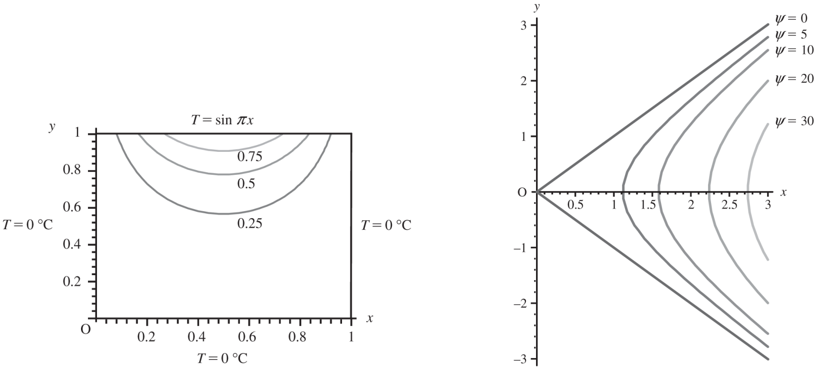

Figure: (left) Variation of temperature on a metal plate $T = T(x, y)$ with fixed temperature ($T = 0$ along 3 edges, and $T = \sin \pi x$ along the top edge). (right) Streamlines of a fluid flow around a corner, described by $\phi(x, y) = 4(x^2 - y^2)$.

Three-dimensional graphs

We can also plot the previous temperature example in 3-dimensions, whereby the height of the plot indicates the value of z.

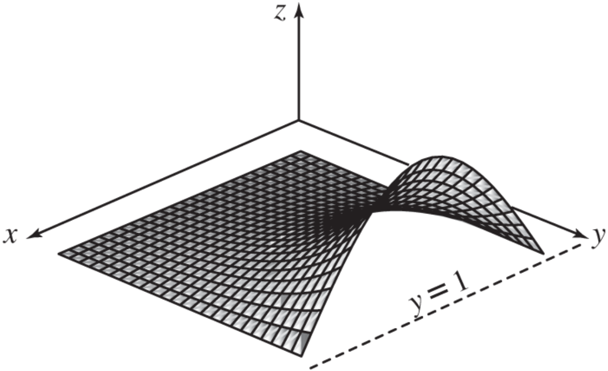

Figure: The temperature distribution from before in 3D.

Introduction

Consider $z = f(x, y)$.

- z is the dependent variable, x and y are the independent variables.
- z can be differentiated with respect to (wrt) x to produce a derivative.
- z can also be differentiated wrt y to produce a different derivative.
- for functions of more than variable we refer to **partial differentiation**.
- instead of writing d we write ∂.

Partial derivatives

If $z = f(x, y)$ then

- $\frac{\partial z}{\partial x}$ represents the derivative of z wrt x, or z_x.
- $\frac{\partial z}{\partial y}$ represents the derivative of z wrt y, or z_y.

Partial differentiation wrt x

Given the function

$$z = f(x) = 5x + 11 \qquad \Rightarrow \frac{dz}{dx} = 5 \tag{4}$$

where here we know that the derivative of the constant 11 is zero.

Given the function

$$z = f(x, y) = 5x + y \qquad \Rightarrow \frac{\partial z}{\partial x} = 5 \tag{5}$$

We differentiate the variable y **as though it were a constant** i.e. the derivative is zero.

Example: Differentiate $z = x^2 - y^2$ *wrt* x.

If we have x and y multiplied together in the function, it is a bit trickier.

Example: Differentiate wrt x*: (a)* $z = yx^2$, *(b)* $3x^2 \cos y$, *(c)* $x^2 e^y + 4y - 3xy^2$.

Partial derivative wrt x

The (first) partial derivative wrt x of a function $z = f(x, y)$ is denoted by $\frac{\partial z}{\partial x}$ and is calculated by differentiating the function wrt x and treating y as though it were a constant.

Partial differentiation wrt y

Similarly, to differentiate a function $z = f(x, y)$ wrt y, we treat any occurrence of the variable x *as though it were a constant*.

Example: Differentiate the following function wrt y: (a) $z = 3y^4 + 4x^2 + 8$, (b) $z = 3x^2y$, (c) $z = e^{3xy^2}$

Partial derivative wrt y

The (first) partial derivative wrt y of a function $z = f(x, y)$ is denoted by $\frac{\partial z}{\partial y}$ and is calculated by differentiating the function wrt y and treating x as though it were a constant.

Note that you may encounter variables other than x and y.

Example: Consider the function $w = f(s, t) = 3t^7 + 4st + s^2$*. Find* $\frac{\partial w}{\partial t}$ *and* $\frac{\partial w}{\partial s}$*.*

Formal definition of partial derivatives

We can formally define the partial derivatives of a function $f(x, y)$ as

$$\frac{\partial f}{\partial x} = \lim_{\delta x \to 0} \frac{f(x + \delta x, y) - f(x, y)}{\delta x} \tag{6}$$

and

$$\frac{\partial f}{\partial y} = \lim_{\delta y \to 0} \frac{f(x, y + \delta y) - f(x, y)}{\delta y} \tag{7}$$

Sometimes these may be written as

$$\left.\frac{\partial f}{\partial x}\right|_y \qquad \text{and} \qquad \left.\frac{\partial f}{\partial y}\right|_x \tag{8}$$

which specifies explicitly what is being held constant during each derivative.

Partial derivatives requiring the product, quotient and chain rules

The use of product, quotient and chain rules work the same as for one variable.

Product Rule:
Example: Find $\frac{\partial z}{\partial x}$ *and* $\frac{\partial z}{\partial y}$ *when* $z = yxe^{2x}$

Quotient Rule:
Example: Find $\frac{\partial z}{\partial x}$ *and* $\frac{\partial z}{\partial y}$ *when* $z = \frac{ye^x}{x}$

Chain Rule:
Example: Find $\frac{\partial z}{\partial x}$ *and* $\frac{\partial z}{\partial y}$ *when* $z = \sin(3x - y^2)$

Some more examples

Example: Find the first partial derivatives of the following functions

$$r = \sqrt{x^2 + y^2} \tag{9}$$

$$u = \frac{x}{x^2 + y^2} \tag{10}$$

$$z = x \ln(xy) \tag{11}$$

$$\tag{12}$$

Finding higher order derivatives

Second partial derivatives are found by simply differentiating the first partial derivatives; this can be done wrt x or wrt y to obtain various second partial derivatives.

Higher-order derivatives

differentiating $\frac{\partial z}{\partial x}$ wrt x produces $\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial x^2}$

differentiating $\frac{\partial z}{\partial x}$ wrt y produces $\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial y \partial x}$

differentiating $\frac{\partial z}{\partial y}$ wrt x produces $\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial x \partial y}$

differentiating $\frac{\partial z}{\partial y}$ wrt y produces $\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial y^2}$

Second partial derivatives

The **second partial derivatives** of z are

$$\frac{\partial^2 z}{\partial x^2}, \qquad \frac{\partial^2 z}{\partial x \partial y}, \qquad \frac{\partial^2 z}{\partial x \partial y}, \qquad \frac{\partial^2 z}{\partial y^2} \tag{13}$$

These are sometimes written as $z_{xx}, z_{xy}, z_{yx}, z_{yy}$.

Some examples...

Example: Given $z = 3xy^3 - 2xy$*, find all the second partial derivatives of* z*.*
Note: how do the computed values of $\frac{\partial^2 z}{\partial x \partial y}$ and $\frac{\partial^2 z}{\partial y \partial x}$ compare?

Example: Laplace's equation is often used in modelling of the temperature of a fluid that is flowing in an insulated pipe. If T *is the temperature of the fluid then under some simplifying assumptions* T *satisfies Laplace's equation*

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0 \tag{14}$$

Show that $T = k - \ln(x^2 + y^2)$ *is a solution to this equation, where* k *is a constant.*

Maxima, minima and saddle points

When we consider functions which depend on two independent variables (i.e. $z = f(x, y)$, we can also have stationary points.

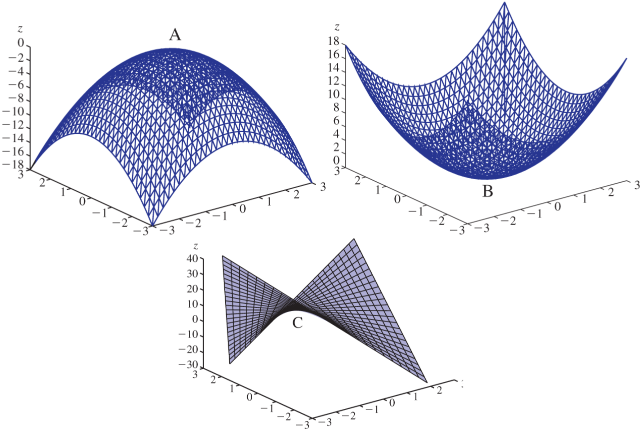

Figure: (left) Maximum point of function $f(x, y) = -x^2 - y^2$; (centre) Minimum point of function $f(x, y) = x^2 + y^2$; (right) Saddle point of function $f(x, y) = 3xy + x + y$.

Finding the location of stationary points

For a function of 2 variables, stationary points occur when *both* partial derivatives are equal to zero.

Stationary point location

Stationary points are located by solving

$$\frac{\partial f}{\partial x} = 0 \qquad \frac{\partial f}{\partial y} = 0 \tag{15}$$

Example: Find the stationary points of the function $f(x, y) = x^2 + xy - 7y$*, and find also the value of the function at the stationary point.*

The nature of stationary points

To find the type of stationary point (maximum, minimum, saddle point), we can do a series of tests using the computed values of the partial derivatives.

Tests to determine nature of stationary points

$$\frac{\partial^2 f}{\partial^2 x}\frac{\partial^2 f}{\partial^2 y} - \left(\frac{\partial^2 f}{\partial x \partial y}\right) < 0 \qquad \Rightarrow \text{saddle point} \tag{16}$$

$$\frac{\partial^2 f}{\partial^2 x}\frac{\partial^2 f}{\partial^2 y} - \left(\frac{\partial^2 f}{\partial x \partial y}\right) > 0 \qquad \text{and} \qquad \frac{\partial^2 f}{\partial^2 x} > 0 \qquad \Rightarrow \text{minimum point} \tag{17}$$

$$\frac{\partial^2 f}{\partial^2 x}\frac{\partial^2 f}{\partial^2 y} - \left(\frac{\partial^2 f}{\partial x \partial y}\right) > 0 \qquad \text{and} \qquad \frac{\partial^2 f}{\partial^2 x} < 0 \qquad \Rightarrow \text{maximum point} \tag{18}$$

Example: Locate and identify the nature of the stationary values of the function $f(x, y) = x^2 + xy - 7y$

1. Function of More than One Variable & Partial Differentiation
   - Function of two independent variables
   - Partial differentiation
   - Higher-order derivatives
   - Stationary values of a function of two variables

2. Integration with more than one variable
   - Multiple Integrals
   - Polar coordinates
   - Multiple integration with polars

Multiple integration

- In the case of functions of more than one variable, just as we can differentiate wrt more than one variable, so we can *integrate over* more than one variable.
- Multiple integration is useful for finding areas and volumes.
- It is also useful for integrating complicated functions over multiple spaces.
- As with partial differentiation, we integrate over variables one and a time, holding all other variables constant.

Summation in 2D

Consider trying to find the area of the rectangle *abcd*

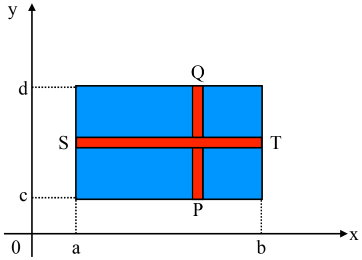

Figure: Integrating to find the area of the rectangle *abcd*.

The element of area is given by

$$\delta a = \delta x \delta y \tag{19}$$

The total area is therefore found by summing-up all of these contributions

$$A = \sum \delta a \tag{20}$$

Summation in 2D

The area of *just* the vertical strip is found by

$$\delta A = \sum_{y=c}^{y=d} \delta x \delta y \tag{21}$$

where δx is held constant. Then, all vertical strips (from $x = a \rightarrow x = b$) are summed-up to find the total area

$$A = \sum_{x=a}^{x=b} \left[\sum_{y=c}^{y=d} \delta x \delta y \right] = \sum_{x=a}^{x=b} \sum_{y=c}^{y=d} \delta x \delta y \tag{22}$$

In the limit that $\delta x \rightarrow 0$ and $\delta y \rightarrow 0$ the summation tends to an integral

Integration over area in 2D

$$A = \int_{x=a}^{x=b} \int_{y=c}^{y=d} dydx \tag{23}$$

Summation in 2D

To evaluate a multiple integral of the form

$$A = \int_{x=a}^{x=b} \int_{y=c}^{y=d} dydx \tag{24}$$

we:

1. work from the inside-out;
2. treat all other variables as constants when not integrating wrt them

$$A = \int_{x=a}^{x=b} \int_{y=c}^{y=d} dydx = \int_{x=a}^{x=b} [y]_c^d \, dx = \int_{x=a}^{x=b} (d-c)dx = (d-c)\,[x]_a^b = (d-c)(b-a) \tag{25}$$

Note that the derivation of the integral A could be been done in the reverse order, summing over horizontal strips first, then adding these together - the order in which we perform the summations does not matter:

Integration order

$$A = \int_{x=a}^{x=b} \int_{y=c}^{y=d} dydx = \int_{y=c}^{y=d} \int_{x=a}^{x=b} dxdy \tag{26}$$

Double integrals

We can extend the ideas of summing over areas in 2D to integration of functions in 2D space.

Double integrals

The expression

$$\int_{y_1}^{y_2}\int_{x_1}^{x_2} f(x,y)dxdy \qquad (27)$$

is called the **double integral** of the function $f(x,y)$ over the $xy-$domain specified:

- $f(x,y)$ is first integrated wrt x (holding y constant) between the limits of $x = x_1$ and $x = x_2$.
- the result is the integrated wrt y between the limits $y = y_1$ and $y = y_2$.

Example: Evaluate $I = \int_1^2 \int_2^4 (x+2y)dxdy$.

Example: Evaluate $I = \int_1^2 \int_0^\pi (4+\sin\theta)d\theta dr$.

Triple integrals

We often have to deal with **triple integrals**, for example, when integrating over 3D space.

Triple Integrals

$$I = \int_{z_1}^{z_2} \int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x, y, z) dxdydz \tag{28}$$

Again, we start inside and work outward (i.e. integrate wrt x, then y, then z).

Note that multiple integrals can be written in an alternative way

$$\int_{z_1}^{z_2} \int_{y_1}^{x_2} \int_{x_1}^{x_2} f(x, y, z) dxdydz \longrightarrow \int_{z_1}^{z_2} dz \int_{y_1}^{y_2} dy \int_{x_1}^{x_2} f(x, y, z) dx \tag{29}$$

Non-constant limits

Consider the case in which we are not integrating over a fixed rectangle

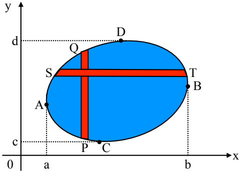

Figure: Integrating to find the area bounded by a closed curve.

where the region R is defined by

$$y_1(x) \quad \leq y \leq y_2(x) \qquad \text{for } a \leq x \leq b \tag{30}$$
$$x_1(y) \quad \leq x \leq x_2(y) \qquad \text{for } c \leq y \leq d \tag{31}$$
$$\tag{32}$$

The double integration (to evaluate the area of R) becomes

Double integration for non-constant limits

$$I = \int_{x=a}^{x=b} dx \int_{y=y_1(x)}^{y=y_2(x)} f(x,y)dy \tag{33}$$

Non-constant limits

Example: Find the area bounded by the curve $y = \frac{4x}{5}$*, the x-axis and the ordinate* $x = 5$*.*

Example: Find the area under the curve $y = 4\sin\frac{x}{2}$ *between* $x = \frac{\pi}{3}$ *and* $x = \pi$ *using double integration.*

Conceptual example of multiple integration

Conceptually, the best example of multiple integration is to consider an integration over a *volume* i.e. over all space in x, y, z.

Consider the density (ρ) of chlorine in a swimming pool (units molecules per unit m).
The density varies at different points in the pool i.e. it varies in space such that ρ is a function of space.
The specific way in which it varies is given by the relation

$$\rho(x, y, z) = 2xy^2z \tag{34}$$

What is the total number of molecules in the swimming pool of size $x = 0 \rightarrow 4$ m, $y = 0 \rightarrow 1$ m, $z = 0 \rightarrow 3$ m?

Polar coordinates in 2D (circular polars)

We are already familiar (from Complex Numbers) with being able to represent the position of a point by either:

1. Cartesian coordinates, x, y
2. Polar coordinates, r, θ

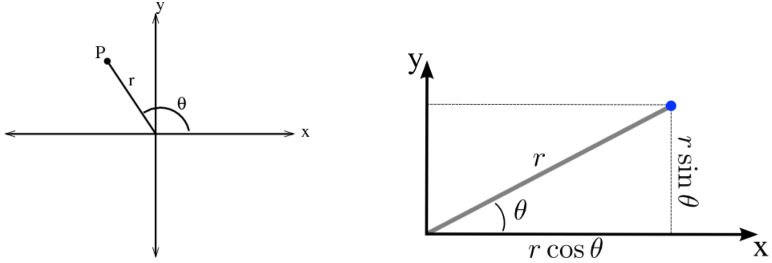

Figure: (left) Position of a point in space defined by polar coordinates; (right) Relation of polar coordinates to Cartesian coordinates.

Relation of polar coordinates to Cartesians

$$x = r\cos\theta \qquad y = r\sin\theta \qquad \tan\theta = \frac{y}{x} = \left(\frac{r\sin\theta}{r\cos\theta}\right) \qquad r = \sqrt{x^2+y^2} \tag{35}$$

Polar curves

Just as we can define a curve in xy-space, where

$$y = f(x) \tag{36}$$

so too can we define a curve in polar coordinate $r\theta$ space where

$$r = f(\theta) \tag{37}$$

Such curves can often be sketched on either on polar coordinate axes, or just on Cartesian xy-axes, recalling that

$$x = r\cos\theta \qquad\qquad y = r\sin\theta \tag{38}$$

Example: Sketch the curves $r(\theta) = 1$, $r(\theta) = \theta$ and $r(\theta) = \sin\theta$ in both Cartesian and polar space.

Polar coordinates in 3D (spherical polars)

θ

- Recall that in 2D, θ represented the angle between the r-vector and the x-axis.
- In 3D, θ now represents the angle between the r-vector and the z-axis.

ψ

- In 3D, we introduce another independent variable, termed ψ.
- ψ represents the angle between the *projection* of the r-vector onto the xy-plane and the x-axis.

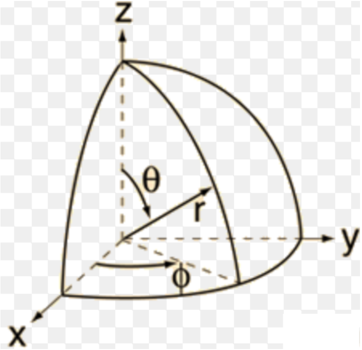

Figure: Definition of spherical polar coordinates relative to Cartesian space.

Integration area in polars

In 2D polar coordinates (r, θ), the element of area is formed between $r \to r + \delta r$ and $\theta \to \theta + \delta\theta$.

Given the arc length subtended by $\delta\theta$ is

$$\text{arc length} = r\delta\theta \tag{39}$$

the area is $r\delta r\delta\theta$.

Elemental area in polar coordinates

$$dxdy \quad \to \quad rdrd\theta \tag{40}$$

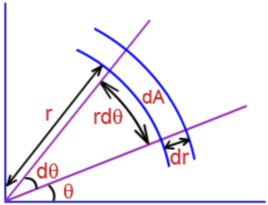

Figure: Elemental area defined in polar coordinates.

Multiple integration with polars

In 2D polar coordinate space, multiple integration is defined as

Multiple integration in 2D polar coordinate space

$$I = \int \int_R f(r,\theta) r dr d\theta \tag{41}$$

To integrate over all space, the integral limits are usually

$$r \ : \ 0 \rightarrow \infty \qquad\qquad \theta \ : \ 0 \rightarrow 2\pi \tag{42}$$

Example: Use polar coordinates to evaluate the integral $\int_{-\infty}^{\infty} e^{-x^2} dx$

Multiple integration with spherical polars

In spherical polars the element of volume becomes

Element of volume in polar coordinates

$$dxdydz \qquad \rightarrow \qquad r^2 \sin\theta drd\theta d\psi \tag{43}$$

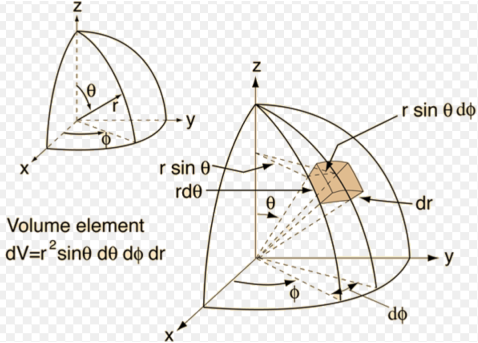

Figure: Elemental volume defined in polar coordinates.

Multiple integration with spherical polars

Such that in 3D polar coordinates (spherical polars), multiple integration is defined as

Multiple integration in 3D polar coordinate space

$$I = \int \int \int_R f(r, \theta, \psi) r^2 \sin\theta dr d\theta d\psi \tag{44}$$

To integrate over all space, the integral limits are usually

$$r \ : \ 0 \rightarrow \infty \qquad \theta \ : \ 0 \rightarrow \pi \qquad \psi \ : \ 0 \rightarrow 2\pi \tag{45}$$

Example: Use spherical polars to integrate the following function $f(x, y, z) = 4z$ over the domain of the upper-half of the sphere defined by $x^2 + y^2 + z^2 = 1$.

What we learnt today

- to be able to represent a function of more than one variable in space and in different coordinate systems
- how to differentiate a function of more than one variable
- how to integrate functions over more than one variable
- how to perform differentiation and integration of function of more than one variable in different coordinate systems, including polar coordinates